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Testing the Dialect Mixture Hypothesis:
Founding Population and Topolectal Variation in Quebec French

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Poster & technical handout

1 Origins of regional variation

- Pronunciation in Quebec French varies across the territory; its main geolinguistic feature is a divide between western and eastern Quebec. At least two hypotheses compete to explain this pattern.
- **1. Inheritance from French regions:** the main settlement hubs drew settlers from different regions of France; linguistic differences across Quebec would continue those between the settlers' home regions (Rivard 1914; Poirier 1994).
- **2. Emergence in the colony:** each settlement hub quickly developed its own local variety from a unique mixture of its pioneers' usages (Trudgill 2004); regional variation would reflect how these early varieties spread through the colony (migration, contact, etc.), not necessarily where in France their founding speakers came from.
- **Did the particular mix of French origins in each settlement hub determine the phonetic variation recorded in the 20th century?**

2 Data & model

1 Founding input



Origins in France — SOURCE

French birthplaces of 3,796 founding settlers (*Fichier Origine*). Their dialects are not observed directly: settlers born close together are assumed to have brought more similar speech patterns.

fixed — similarity halves at 100 km, keeps declining

2 Emergence of new varieties of French in the St. Lawrence valley



First settlements — SOURCE

Parishes founded before 1763 are modelled as blends of their pioneers' inputs. Quebec City, Trois-Rivières, Montreal, etc. all drew on somewhat different mixtures.

fixed — pioneers: arrived in the parish's first 50 years

estimated — weight of this source

3 Geolinguistic development from the first settlements, generation after generation



Migration — CHANNEL

Speech patterns travel with people: actual moves between parishes are reconstructed from life trajectories (PRDH; Dillon et al. 2018).

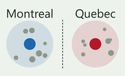
fixed — computed for every generation



Neighbour contact — CHANNEL

Parishes adopt some of each other's speech patterns, more from larger and closer ones (*gravity model;* Trudgill 1974).

tested — 0–20% per generation; reach 5–25 km



Spheres of influence — CHANNEL

Each parish adopts some of the speech patterns of the dominant city nearby, Quebec City or Montreal, weighted by size and distance.

tested — 0–20% per generation; reach 50–200 km



Local innovation — SOURCE

Each generation, every parish develops small innovations of its own and drifts slightly from the others; Quebec City (colonial capital) innovates more, until 1763.

fixed — Quebec City ×10 until 1763

estimated — weight of this source

HOW THE MODEL IS TESTED

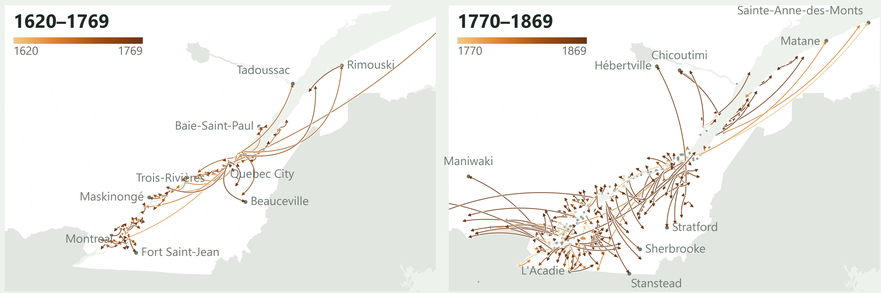
Ten consecutive 25-year generations (1620–1869) yield a predicted phonetic distance between every pair of the 125 localities of the *Atlas linguistique de l'Est du Canada* (ALEC; Dulong & Bergeron 1980). Predictions are compared with the observed distances (corrected for fieldworker biases†), by spatial cross-validation: each of 12 regional blocks is predicted by a model whose weights and tested values are chosen on the other 11 only.

Full model specification in the handout. Exploratory model.

† Saint-Amant Lamy (2026)

3 Settlement history 1620–1869

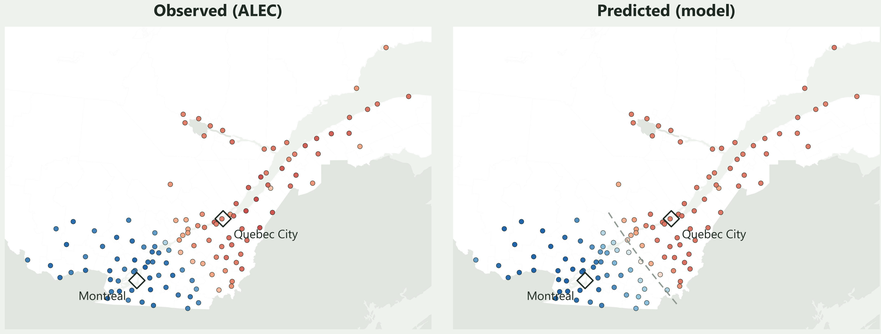
Settlement patterns reconstructed from PRDH life trajectories (places of baptism, marriage, burial), not simulated.



→ **main origin of each new locality's founders** ● localities already settled

4 Predicted vs. observed variation

Predicted map: *the model of section 2 run through the settlement history of section 3.*





Colour = main dimension of variation in the phonetic distances between localities (classical MDS), not a single phonetic feature.
Similar colours = similar overall pronunciation. Both maps use the same scale.
Dashed line (predicted map) is the final boundary between the Montreal and Quebec City spheres of influence.

ALL COMBINATIONS, FIXED IN ADVANCE (R^2 out of sample)

Sources of variation \ *Channels of diffusion*	Settlement only	Neighbour contact	Spheres of influence	Both channels
Founding settlers	0.43	0.65	0.70	0.77
Local innovation	0.13	0.24	0.41	0.45
Both sources	0.43	0.65	0.70	0.77

Reading the table: adding each diffusion channel improves the fit; both together do best. Adding local innovation to the founding settlers changes almost nothing (bottom row = top row; estimated weight 2%); on its own, innovation reaches only 0.45.

RETAINED MODEL (gold frame)

R^2 0.77 out of sample

78% of estimated explainable variance (R^2 ceiling: 0.99)

Source: founding settlers + effect of innovation
Channels: migration + contact + spheres of influence

Details

fixed — settlement: observed migration patterns

estimated — both sources: settlers 98%, innovation 2%

tested — contact: 20%/generation, 25 km*

tested — spheres: 20%/generation, 200 km*

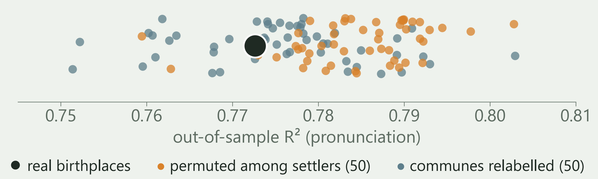
(Other designs of the channels: R^2 0.75–0.78, same retained values)

* Upper end of tested values, selected independently in each of the 12 folds (nested cross-validation); not estimates of historical contact rates.

5 Testing French origins

Same model, rerun 100 times with French birthplaces reassigned at random; everything that happened in Quebec is kept. If French regions really matter, real birthplaces should predict the ALEC map better.

- **50 permutations:** birthplaces are shuffled among individual settlers.
- **50 relabellings:** the locations of communes in France are shuffled; settlers from the same commune stay together (the most direct test of French geography).



- **Result:** real birthplaces yield no better predictions than randomly reassigned ones (R^2 0.77 vs. 0.75–0.80).
- **Mechanism:** random birthplaces yield almost the same predicted map ($r \approx 0.99$). The model suggests that the map primarily reflects an early difference between Quebec City and Montreal, whatever its source, subsequently propagated through settlement and contact. Local innovation alone produces a much poorer fit (R^2 0.45).
- **Interpretation:** each hub quickly formed its own slightly distinct variety by selecting among the many speech patterns its pioneers brought; their French origins shaped the pool of available patterns, but do not detectably predict which ones prevailed in each hub.

Imported ingredients, home-brewed map.

Quebec French was built largely from linguistic features brought from France, but its geolinguistic structures seem to originate within the colony itself: local varieties formed in the first hubs and spread along settlement and contact networks.

6 Conclusions

- **No detectable gain from French regions:** real birthplaces predict no better than random ones
- **Founders, settlement history and contact explain most of the variation:** 78% of the estimated explainable variance
- **A plausible scenario:** slightly distinct norms formed early in Quebec City and in Montreal, then spread through settlement, contact and their spheres of influence; the east–west divide lies where these spheres meet
- **Compatible with new-dialect formation** by mixing and levelling, the process typical of colonial varieties (Trudgill 2004)

Limits:

- *French dialects are not observed directly (birthplace proximity as a proxy)*
- *Aggregate distances between localities, not individual linguistic features*
- *A weak regional effect cannot be ruled out*
- *The simulation stops in 1869, before the ALEC speakers were born*
- *Exploratory model*

References: Full list in the handout (QR code, top right).

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